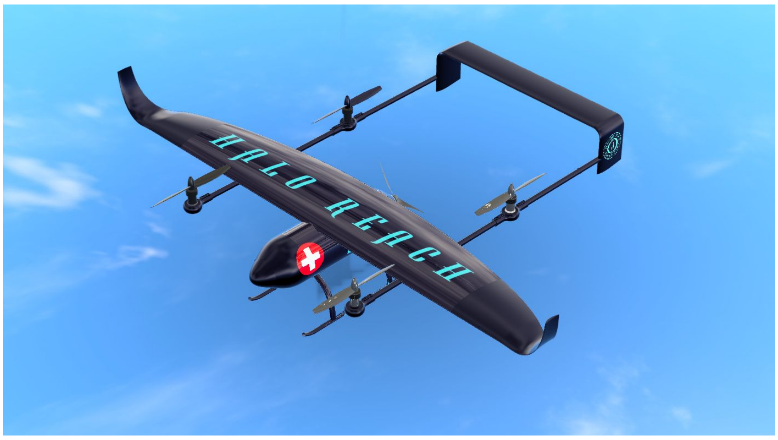
HALO REACH: Propulsion System Design

UAS for Disaster Relief - AE3 2024



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Abstract

In light of the devastating earthquakes in Turkey/Syria and Morocco, which killed 60,750 and 3000 respectively [1][2], it is evident that the current solutions to rescue the victims of natural disasters are ineffective. This project aims to design an Unmanned Aerial System (UAS) which can improve the efficiency and effectiveness of disaster relief efforts. When it comes to natural distasters, the stakes are high, and any additional lives saved can be the difference needed for a family to stay whole.

This report outlines the design process with a focus on the systems and propulsion of the UAS. Covering the conceptual design, preliminary power analyses, any choices made and their justifications, and finally a reflection and discussion of further improvements.

The results show that a UAS can be feasibly developed with the capabilities required to significantly aid in the rescue efforts following a natural disaster. This will be done through fully autonomous delivery of essential relief supplies, or high quality surveillance and providing communication with emergency services to the victims. This innovative approach shows promise in improving the overall effectiveness of disaster relief to save more lives.

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Nomenclature

η	Efficiency (%)
ρ	Air density (kg/m^3)
A	Rotor blade plan area (m^2)
C_d	Coefficient of drag (-)
D	Drag (N)
E	Endurance (s)
g	Acceleration due to gravity (N/kg)
L	Lift (N)
m	Mass (kg)
P	Power (W)
R	Range (m)
S_{ref}	Aircraft reference area (m^2)
T	Thrust (N)
t	Time (s)
V	Velocity (m/s)
W	Weight (N)
BEM	Blade Element Momentum
HALO	Humanitarian Aid Logistics Operations
ICE	Internal Combustion Egnine
MTOW	Maximum Take-off Weight
REACH	Rapid Emergency Aid and Communications Hub
SFC	Specific Fuel Consumption
UAS	Uncrewed Aerial System
UAV	Uncrewed Aerial Vehicle
VTOL	Vertical Take-Off and Landing

1 Introduction

As aerial technology is constantly improving, new areas where this technology can be applied are being found; This project will discuss the feasibility of an application in natural disaster scenarios.

1.1 Context and Motivation

The case study of the recent earthquake in Turkey/Syria, which resulted in the death of 60,750 people [1], desperately highlights the need for more effective disaster relief solutions. Current methods have been seen to be ineffective in providing timely and efficient rescues; in a disaster scenario, the ability to quickly deliver aid is crucial, as every minute wasted could potentially be another life lost.

Looking deeper into the factors that determine casualty levels in earthquakes, it can be noted that many of the deaths occur not at the time of the disaster, but in the aftermath of it; such as in the case of a building collapse. Figure 3 shows an estimation of mortality of trapped occupants of a collapsed building, as time increases post collapse, M4 is the proportion that unfortunately die immediately on collapse, while M5 is the proportion that are announced deceased once rescue efforts in that building cease. About 20% of occupants die instantaneously, and only about 20% are rescued. This leaves an unacceptable 60% that survived the disaster itself, but died later due to failures of the rescue methods. [3]

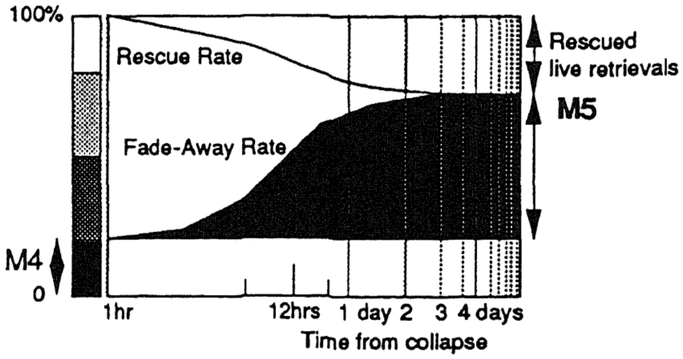

Figure 3. % of the injured that subsequently die before they can be rescued [3]

The shortcomings of current methods of disaster relief can be constituted by the many problems that are encountered; including, but not limited to: lack of preparedness, of both the victims of the disaster, and the emergency services; lack of coordination and communication within the emergency services; deficiencies of equipment and supplies, especially in poorer countries; and finally, access to rural or damaged areas to provide equipment or ambulances can pose a challenge.

1.2 The objective

This project aims to address these shortcomings that are negatively affecting the current disaster relief methods; through the design of an Unmanned Aerial System (UAS), the critical issues of disaster relief can be mitigated.

This will be done by providing rapid, autonomous delivery of essential supplies, performing high quality surveillance for rapid assessment, and facilitating a reliable means of communication between emergency services, medics, and the victims.

By successfully creating a solution to these challenges, this project hopes to improve the speed and effectiveness of disaster relief operations, and save lives.

2 Conceptual design

The conceptual design phase is the first step in the design process, where the major decisions, definitions and constraints are made. This dictates the direction of the project, and provides a foundation for the team to then delve deeper in the design.

2.1 Baseline configuration

Choosing a baseline configuration for the Unmanned Aerial Vehicle (UAV), named ”REACH”, was a highly collaborative process, both within the Systems and Propulsion sub-team and the project team as a whole. Under direction of the project coordinators, each sub-team was tasked with pitching their preferred configuration to the other sub-teams, which they would then be questioned and have to justify their choice. This peer scrutinisation ensured that any choices made were well thought out.

The Systems and Propulsion sub-team approached choosing their preferred baseline configuration through researching the current market of UAVs and creating a list of ”Pros and Cons” of each configuration, which can be found and further explained in Tharm's report [4]. The sub-team decided to pitch a ”tilt-wing” configuration; where take-off and landing would function like a multi-rotor UAV, then transitioning into fixed-wing flight by rotating the wings with the propulsion system attached to them. This was chosen for its theoretical high efficiency, using the same propulsion system for both the VTOL and fixed-wing flight. Further, as a requirement, REACH must be able to fit within a 6-by-6 m square [5], if wings were to be used, the wingspan would typically be constrained to 6 m, however, by rotating them, that constraint could be bypassed.

This idea was ultimately rejected by the wider team due to the complexity of the moving parts, which would cause great difficulty for the structures and flight dynamics sub-teams to do their sections within time constraints for the project. Overall, a vote and trade-off calculation was done as Hasan's report expands on [6], the team decided on a ”lift and cruise” configuration. This will consist of a fixed-wing UAV with a forward facing propeller, and a separate set of upward facing rotors for VTOL capabilities. This design will be less efficient than other options, due to the added mass of extra propulsion systems, and added drag of the unused rotors during fixed-wing flight, but these cons are heavily outweighed by the simplicity of the design. While efficiency is an important factor to consider, simplicity, reliability and robustness are all properties that take precedence when it comes to this project's objectives. A an image of the model of REACH, showing the baseline configuration, can be found on the title page.[7]

2.2 Options of power supply

There are many factors to take into consideration and discuss when choosing the source of energy for REACH:

2.2.1 Energy density, J/kg

This is one of the most crucial parameters to consider, as it directly influences the performance characteristics with pre-defined requirements such as range and endurance [5]. A higher energy density allows more energy storage with less of a weight penalty and thus more efficient flight, which is an extremely desireable feature. Figure 4 shows a chart of various fuels and their energy densities, it can be seen that there is a difference of two orders of magnitude between lithium batteries and hydrocarbons like Jet-A, while hydrogen is another order of magnitude above hydrocarbons.

From this information, hydrogen power seems like an obvious choice; however, there are many other operational factors to consider, as discussed below.

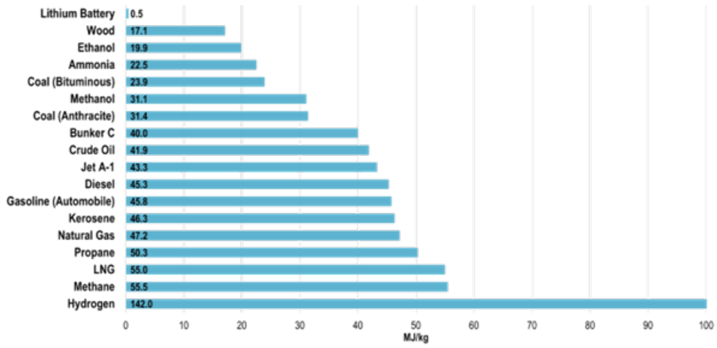

Figure 4. Energy density of various fuels [8]

2.2.2 Volumetric density, kg/m^3

REACH has a requirement of fitting within a 6-by-6 metre square for storage [5], this means that the fuel chosen cannot have too large of a volume requirement. Table 2 lists a few fuels that could be used to power a UAV, this shows where hydrogen falls short; in its most dense, usable state (liquid), its density is far too low; while it would require the smallest mass to power the mission, it would also have the largest volume requirement, and as this is heavily constrained, hydrogen becomes less justifiable as an option. Batteries, however, have a significantly higher density, so while there will be a larger weight penalty as discussed previously, very little volume will be needed, which allows more space for other systems and structure.[9][10]

Fuel name	**Density** $[kg/m^3]$
Lithium ion battery	2500
Jet-A	800
Liquid Hydrogen	70

Table 2. Mass/volume density of some fuels [9][10]

2.2.3 Storage methods

In some cases, the means of storing the fuel can cause further issues. This, again, is an aspect where hydrogen falls short. In order to store hydrogen in its most dense state, it must be a cryogenic liquid; storing a cryogen on an aircraft presents large challenges. The high pressure requires heavily reinforced containers, adding a mass penalty; maintaining cryogenic temperatures requires a lot of insulation material that take extra volume; liquid cryogen storage must be allowed to bleed, otherwise it becomes dangerously explosive, so the fuel storage becomes inefficient and must be oversized. This was the "final nail in the coffin" for hydrogen as a choice, and it was rejected. However, in this aspect, batteries and liquid hydrocarbons are great; they are very simple to store and use, batteries can be directly placed within the fuselage, and a simple fuel tank is used for liquid hydrocarbons [11].

2.2.4 Environmental considerations

This is the effect of the environment on the power source, but also the power source's effect on the environment. This project's objective means that the power source must be robust, operating in a wide variety of potentially extreme environments. Some desireable features would include: a wide range of operating temperatures, pH levels and altitudes. Batteries and liquid hydrocarbons tend to be the best options in this case, Ishaan's report [12] investigates further on this topic.

Sustainability and environmental impact are important considerations, especially given this project's mission, it would be ironic to aid in saving people from natural disasters, while actively contributing to the worsening of the climate that causes them. However, as will be discussed later in the report, this may be a necessary sacrifice to ensure the success of the project.

2.2.5 Discussion

This is not an exhaustive list, but discusses the most important factors in the decision. As per the briefing [5], REACH must be able to use batteries or Jet-A (or both) as a store of energy for the propulsion system. This constraint left the team with a choice between a fully electric system, or a hybrid system that uses both batteries and Jet-A; however, upon consulting with Dr. Fasel, a third option was presented, one of a fully Jet-A fueled UAV, where Jet-A is used to spin the rotors [13]. While this has been demonstrated to work for a quad-copter and could theoretically be applied to this project, increasing mass efficiency, it is still a very novel technology and the methods are not currently being shared. Perhaps, towards the end of the lifecycle of the project, a refurbishment could be considered where this technology would then be applied to REACH.

Thus, at this stage in the conceptual design, the team moved forward with the remaining two possibilities, both were to be considered, and an investigation into the feasibility of both options would provide insight for a justified decision of one over the other.

2.3 Initial weight sizing and power source decision

Research was done into other similar UAVs with either similar baseline configurations or similar performance requirements, looking into categorising them as either ”fully electric” or ”hybrid”. This allows some trends to be found, and projected to find an estimate for this design. As per the briefing [5], a maximum take-off weight (MTOW) of 160 kg is required, so by finding a historical relationship between MTOW and empty weight of UAVs, an initial estimate of the weight breakdown could be found. Figure 5 shows the trendlines found from the various UAVs.[1] It can be noted that the trendline for the hybrid systems has a noticably shallower gradient than that of fully electric systems. This means that for a given MTOW, the estimated empty weight will be lower for a hybrid system when compared to fully electric.

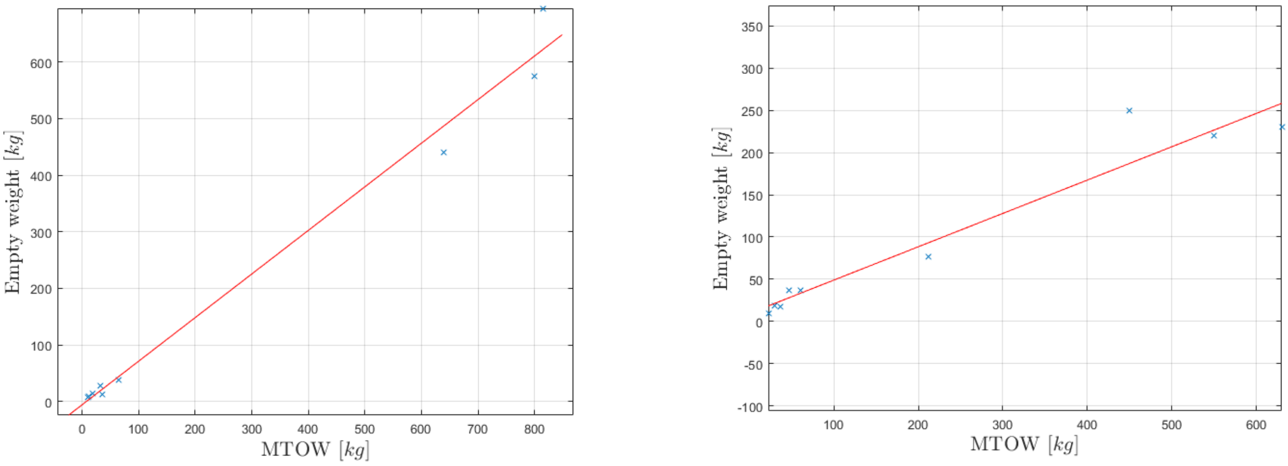

Figure 5. fully electric (left) and hybrid (right) propulsion systems empty weight vs MTOW trends

The required payload masses for the different mission profiles are also defined in the briefing [5]; the team believed that the most constraining case for fuel would be the long endurance mission, this was later found to be wrong, but for initial sizing, the long endurance payload mass of 20 kg was used. From there, along with the historical trendlines, a very rough initial mass breakdown of REACH can be found. Table 3 shows these values.

[1] Full list of references for Figure 5: [14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32]

	MTOW = 160 kg	
Propulsion system	**Empty mass** $[kg]$	**Energy storage mass** $[kg]$
Hybrid	80	60
Fully electric	100	40

Table 3. Initial weight breakdown estimations of a hybrid and fully electric system

This provided an ”upper bound” to be used for a feasibility analysis. If an estimate of required energy storage mass (batteries or batteries plus fuel) is found to be above the allowance, that choice is not feasible and can be rejected.

2.3.1 Fully electric

Battery technology is constantly improving, while the current state of batteries' energy density is comparatively low, the lifecycle of this project is expected to extend significantly into the future as Patricia's report [33] expands on, so it can be expected that the energy density of batteries will improve over time as the UAVs begin to be manufactured.

Operating under the assumption of constant mass, which is valid for a fully electric system, a simple analysis was done, using battery energy density as an independent variable, and finding how much battery mass is required to complete the take-off, landing, cruise and loiter phases. This neglected the transitions and short climbs, which, for an initial estimation, would not be too significant. Blade Element Momentum (BEM) theory was used to estimate power requirements for VTOL, which is explained in depth in Tharm's report [4]. For the fixed-wing segments, a force balance approach was taken, assuming steady, level flight; thrust, T, lift, L, drag, D and weight, W, must all balance each other, and an estimate of thrust can be found:

$$T = \frac{W}{L/D} \tag{1}$$

Values of lift-to-drag ratio and airspeed, V, in cruise and loiter, were approximated and provided by the aerodynamics sub-team [34]. Using a definition of power, P, noting that this is the useful power output, not what is drawn from the battery, a propulsive efficiency, η_p, of 0.8 is applied as an empirical value for an efficient UAV [35], and substituting in the relation for Thrust:

$$P \triangleq \frac{V \times T}{\eta_p} \tag{2}$$

$$\therefore P = \frac{V}{\eta_p} \frac{W}{L/D} \tag{3}$$

This provided an estimate of power requirement, and the duration of these segments are defined in the briefing as 1.25 hour cruise, and 10 hour loiter [5]. The total energy storage requirements can be found by simply multiplying the power requirement, which would be constant due to the constant weight, by the time in the segment, t_i, and summing this over the N different segments. Battery mass is then found through the independent energy density, Figure 6 plots this relationship.

$$Battery\ Mass = \frac{\sum_{i=1}^{N} \left[t_i \left(\frac{V}{\eta_p} \frac{W}{L/D} \right)_i \right]}{Energy\ density} \quad (4)$$

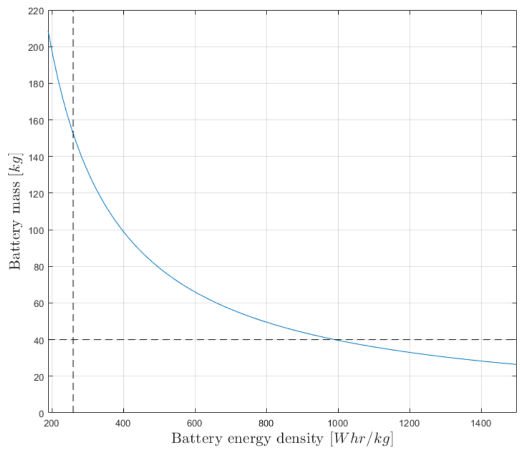

Figure 6. Battery mass requirements as energy density varies

Shown by vertical dashed line in Figure 6, the best current lithium-ion batteries available tend to be around 260 Whr/kg [10], and the horizontal dashed line shows the predicted mass allowance for batteries from Table 3. For a fully electric system to be considered feasible, the curve must be below where the two dashed lines meet, this would only be the case if the battery energy density is above 1000 Whr/kg, and through some research, the state of the art of batteries, that are currently being developed, only go up 500 Whr/kg [36]. An alternative would be to find a way to bring the curve down, which in accordance with Equation 4, could be done by reducing the time or the power requirement for each segment, however, these unfortunately are mostly constrained [5]. This means that a fully electric system is simply not feasible for the desired mission, currently and for the near foreseeable future.

2.3.2 Hybrid

Having learnt that a fully electric system would not be possible, hopefully a hybrid system would be the solution for completing the mission. For this propulsive configuration, an engine burning Jet-A would be used for fixed-wing flight, while motors powered by batteries would be used to power the rotors for VTOL capabilities.

Still assuming steady, level flight for the cruise and loiter segments, the Breguet range/endurance equations could be used to estimate each segment's weight fraction [37], $\frac{W_i}{W_{i-1}}$, through an estimated specific fuel consumption [38], SFC, of 0.4 $kg/kWhr$, the desired range or endurance, R or E, and other known flight parameters.

$$\text{Range: } \frac{W_i}{W_{i-1}} = e^{\left(\frac{-R \times SFC}{V \times (L/D)}\right)} \quad (5)$$

$$\text{Endurance: } \frac{W_i}{W_{i-1}} = e^{\left(\frac{-E \times SFC}{L/D}\right)} \quad (6)$$

This provided an estimated 34 kg of Jet-A required to complete the mission, which, with the predicted upper bound of available mass for energy storage of 60 kg, leaves 26 kg for batteries to power the VTOL rotors. This was a very promising result, and the sub-team decided to move forward with the design of a hybrid system.

The choice of a hybrid system means that a sacrifice must be made in the sustainability and environmental impact of REACH as Jet-A will be burnt. However, it also presents some added benefits, such as an excellent suggestion made by one of the team's academic supervisors, which was to add a generator connected to the engine, gaining the ability to recharge the batteries during flight, at the cost of some extra fuel. This could significantly reduce the required battery mass, and will be further investigated later in the report.

2.4 Powerplant selection

In order to progress into the preliminary analysis for propulsion systems, powerplant selection had to be done, so that exact engine, battery, and motor specifications could be used for accurate results. The choice of motors to spin the VTOL rotors was based off the power output requirements for take-off, through the use of BEM theory. While the choice of battery to power the motors (plus the other systems) was based off research of the current state of the art in battery technology. Both of these are explained in depth in Tharm's report [4].

The choice of engine was heavily dictated by the design constraints. It must operate efficiently at the design MTOW of 160 kg and cruise speed of 41.1 m/s. Immediately, the sub-team had to reject any form of turbomachinery; this was because the cruise speed, at the design cruise altitude of 500 m [5], corresponds to a Mach number of 0.12. Figure 7 shows that at this Mach number, all options of turbomachinery are extremely inefficient. They could not be used.

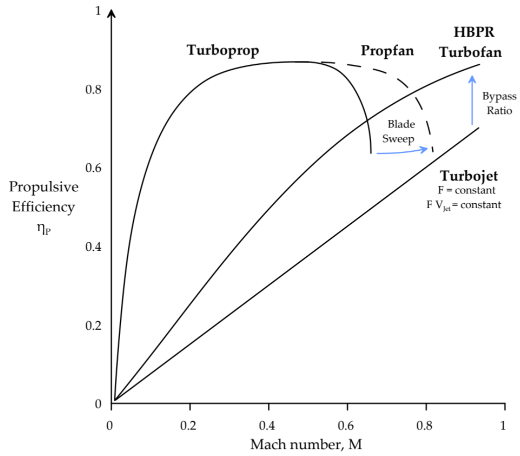

Figure 7. Propulsive efficiency as Mach number varies [39]

After rejecting turbomachinery, this left the sub-team with the option of an internal combustion engine (ICE) that would generate torque to spin a propeller. Here, the choice of ICE was heavily constrained by the requirements defined in the briefing; it must be able to burn Jet-A [5]. This presented a challenge as a typical ICE would be fueled by various forms of petrol or diesel (such as "avgas" for small aviation engines [40]), and understandably so; since the engines go through many combustion cycles per second, the fuel used must be introduced and well mixed with the air in a very short time-frame, which is best done with less dense fuels. Jet-A is kerosene-based, containing some of the longest chain hydrocarbons (10 - 16 carbon atoms per molecule [41]) that can still be used as a liquid fuel, and such, it is comparatively dense. Jet-A is generally not suited for ICEs; however, there are a few to choose from consisting of two main classifications: piston engines or Wankel rotary engines.

Wankel engines have some advantages when compared to a conventional piston engine, such as: a higher power-to-weight ratio, very little vibration, and adaptability to various fuels. However, the Wankel engine also suffers from very poor thermodynamics; the intake air is compressed over a very large surface area, creating a "long, thin, and moving"[42] combustion chamber, this causes poor mixing and slow combustion, meaning unburnt fuel is exhausted. While piston engines are highly reliable with better mixed and more complete combustion, extracting more energy out of the fuel.

Ultimately, a Wankel engine will provide a good solution to burning Jet-A on a small UAV, but at the cost of an increased fuel consumption to account for its inefficiencies[42]. Given REACH's mission of long endurance around disaster zones, if a piston engine can be fit within weight constraints, the increased reliability and fuel efficiency are highly desireable features. A piston engine was found which is designed to burn Jet-A and seemed like a good weight, the **Hirth 4202HF**. Table 5 in Appendix A shows the sub-team's choices of engine, motor, and battery.

3 Preliminary analysis

With the major decisions made and important parameters defined, a more in-depth preliminary analysis could be performed. This section aims to review the initial assumptions, such as the fuel masses from Table 3, and perform an investigation to either verify or update these values. Then, metrics for aircraft performance can be found.

3.1 The mission profiles

As per the briefing [5], REACH must be designed for two main mission profiles: a short supply delivery mission with a payload of 50 kg, and a long endurance communication relay mission with a payload of 20 kg. REACH must be designed for a MTOW of 160 kg in all cases; this presented a challenge as there is a difference in 30 kg of payload between the two mission profiles, in order to maintain a constant 160 kg take-off mass, that difference must be made up somewhere else.

One proposed solution was an external fuel tank of 30 kg that could be "plugged in" for the long endurance mission. This was an interesting idea, it also created an opportunity to revisit the arguement for sustainability; perhaps the less energy demanding supply delivery mission could be performed fully electric, and then when the endurance is needed, an external fuel tank is plugged in, so the engine can be used, which will also balance the difference in payload mass. This would simplify the design of the internal structure, and reduce the mass of fossil fuels burnt, decreasing the environmental impact of REACH. However, an engine had already been chosen and would be embedded within the structure of REACH; if it is not used, it would be "dead-weight" to carry during the supply delivery mission. Also, extra electronic components would be required to spin the propeller instead of the engine. Ultimately, given the complexity of this solution and the time constraints of the project, this idea was rejected in favour of a more conventional approach.

Dr. Fasel offered the simplest solution; oversize the fuel tanks, only partially filling them for the supply delivery mission, and have enough space to add the extra 30 kg of fuel needed to make up the difference in payload mass for the long endurance mission. After consulting with the rest of the team, this option was taken forward.

The next step was to break down the mission profiles into individual segments, and derive general relations of power requirement as a function of time. These are crucial to model accurately, as by plotting these relations one after another for the whole mission profile, the area under the graph will be equal to the total energy used for the mission profile and can directly be used to find accurate values of fuel and battery mass.

3.1.1 Take-off/Hover/Landing

BEM theory was used for these segments, using more up-to-date parameters, this is covered in depth in Tharm's report [4].

3.2 Transition to fixed-wing

The transition from VTOL to fixed-wing flight was modelled as: starting from level hover, full throttle of the engine is applied, and as REACH accelerates forward, lift is increasingly generated by the wings, so the thrust generated by the VTOL rotors is decreased accordingly. This way, the forces in the vertical direction remain balanced, and the UAV maintains a constant altitude while accelerating forward.

BEM theory provides a method of finding power requirements of the motors (Equation 7), based off the thrust produced by the VTOL rotors, T_{VTOL}, air density, ρ, plan area of the rotor blades, A, and dividing by the VTOL efficiency, η_{VTOL}, translates the useful power output of the VTOL rotors to the power of the motors [4]. Further, by the assumptions made for the transition, T_{VTOL} can be related to the lift generated by the wings (Equation 8), where mass and lift are functions of time due to fuel burning and acceleration ($m(t)$ and $L(t)$ respectively), and g is the acceleration due to gravity at the Earth's surface:

$$P_{motors} = \frac{1}{\eta_{VTOL}} \frac{(T_{VTOL})^{1.5}}{\sqrt{2\rho A}} \quad (7) \qquad T_{VTOL} = m(t) \cdot g - L(t) \quad (8)$$

In order to find lift as a function of time, velocity in the horizontal direction as a function of time is needed. A differential equation, relating velocity and time, was formed using Newton's second law, the full derivation can be found in Appendix C.

$$\frac{P_{engine}\,\eta_p}{V} - \frac{1}{2}\rho V^2 S_{ref} C_d = (m_0 - (SFC \cdot P_{engine})\,t\,)\,\frac{dV}{dt} + V\,(-\mathrm{SFC} \cdot P_{engine}) \quad (9)$$

This could be solved through separation of variables, but at this point, the author opted to take a numerical approach using MATLAB's "Symbolic Math Toolbox" [43]; Figure 8 shows the numerical solution, using parameters provided by the Aerodynamics sub-team [44][45]. It is good practice to inspect the results of numerical methods, to ensure that it converges to a sensical solution. It can be seen that: starting at the origin with zero velocity meets the initial condition; an initial rapid acceleration makes sense, reaching cruise speed in 20 seconds, which agrees with Shun's much more complex model that he developed for control [46]; and plateaus at 51.5 m/s as drag catches up with thrust with the given flight parameters. This result looks like it can be trusted.

Taking this result, substitutions are made back into Equation 7, finally providing an accurate model for the power requirements of the motors during transition, and the power output of the engine is known already from the assumption of full throttle. Figure 9 plots this relation; notably, the power requirement reaches zero at 20 seconds, this is because the flight parameters being used are that of cruise, so once cruise speed is reached, the lift generated by the wings is equal to the weight of the UAV, and the VTOL rotors are no longer required. At that point, the throttle would be reduced appropriately and transition would be considered complete.

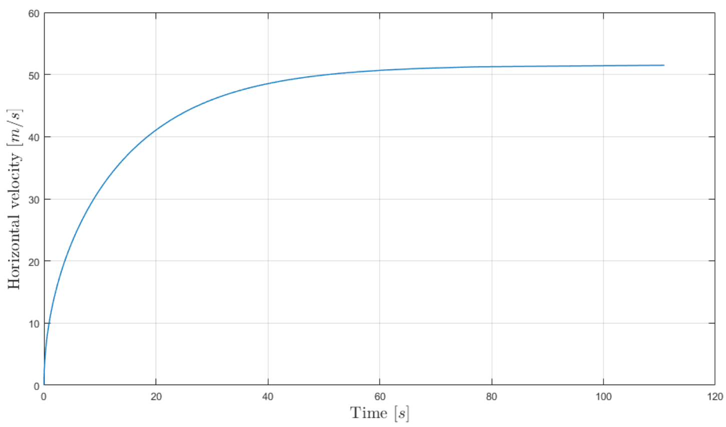

Figure 8. Horizontal velocity during transition to fixed-wing flight

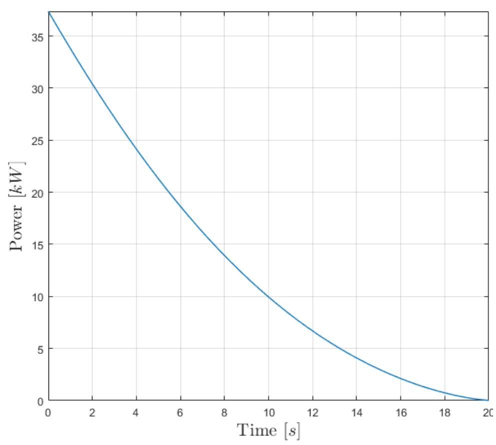

Figure 9. VTOL motor power requirements during transition

3.2.1 Climb/Descent

Values of thrust and airspeed for these segments were calculated by the aerodynamics sub-team [44], the power required by the engine is simply found by the definition in Equation 2

3.2.2 Steady, level flight

This applies to cruise and loiter segments; while there will be a small constant roll angle during loiter, the effects of this are assumed negligible. In steady, level flight, the VTOL rotors are not used at all, and engine characteristics can be used to directly calculate mass as a function of time.

An assumption was made that the power requirement for flight is directly proportional to mass, this was verified through research, finding that the Breguet range equation uses the same assumption [37]. It is known that fuel consumption is proportional to power, with SFC being the constant of proportionality; SFC does change with power output of the engine, however in the small range of power that will be needed for a segment of steady, level flight, it can be assumed approximately constant. Combined, this means that fuel consumption is proportional to mass, forming a very simple differential equation, where K is some constant of proportionality.

$$\frac{dm}{dt} = Km \tag{10}$$

Using general initial conditions, at time, $t = 0 \implies m(0) = m_0$ and $\frac{dm}{dt}(0) = -SFC \cdot P_0$. The solution is trivial, and mass as a function of time is found. For plotting purposes, this was then substituted into Equation 2, to find power as a function of time.

$$m(t) = m_0 \, e^{\left(\frac{-SFC \cdot P_0}{m_0} t\right)} \tag{11}$$

3.2.3 Transition back to VTOL

This is highly complicated, and an accurate model of how this is done is explored in depth in Shun's report [46]. Due to time constraints, simplifying assumptions were made for this analysis; so long as the approach is conservative, it is ensured the battery mass will not be an under-estimate.

In this case, the transition back to VTOL was modelled as: turning off the engines, slowing down with drag, and gradually increasing the thrust of the VTOL rotors to account for the loss of lift, maintaining constant height. The analysis for this took the exact same approach as the transition into fixed-wing flight, starting with Equation 7, and setting T_{prop} to be zero. Figure 10 shows the relationship of velocity with time, starting from loiter conditions, slowing down with only drag. The velocity asymptotes to zero, taking a theoretical infinite time to fully stop; to circumvent this, a "cut-off" velocity was introduced, which is the speed at which the lift generated by the wings is considered negligible, and the multi-rotor dynamics dominate. After consulting with the flight dynamics and control sub-team, 8 m/s seemed reasonable and is shown by a dashed line.

This model of transition takes 100 seconds, shown by a dashed line in Figure 11, which plots the power requirement of the motors during transition. This means that the energy requirement for transition back to VTOL would be the area under the graph, between the bounds of time = [0,100]. Realistically, the transition would not take 100 seconds; the true method of transitioning would involve inducing a pitch angle for thrust vector control [46] and a much quicker stop. This means the results of this analysis will be an overestimate of energy requirements, confirming that this is indeed a conservative approach.

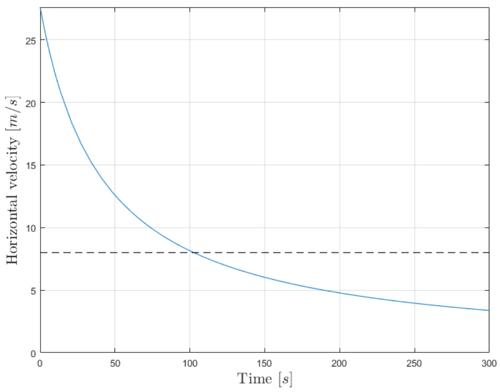

Figure 10. Horizontal velocity during transition to VTOL

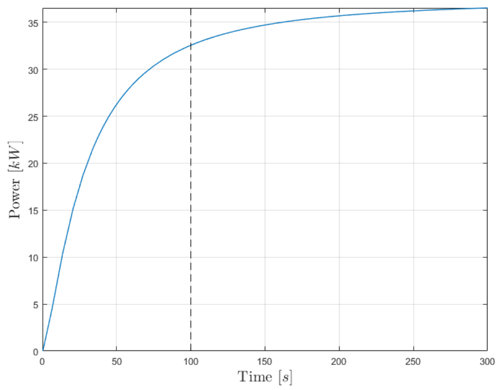

Figure 11. Motor power requirements during transition to VTOL

With this, the mission profile breakdown is complete, and the pieces can be put together. The large plots of power requirements for the most demanding mission profiles with respect to the motors and engine, supply delivery and long endurance respectively, can be found in Appendix D.

3.3 Battery recharging

Following a suggestion discussed in Section 2.3.2, an investigation into the possible benefits of using the engine to recharge the batteries could now be done. Conveniently, the manufacturer of the chosen engine also included an optional 1 kW generator for exactly this purpose [47]. As the motors are only used for take-off, landing, and transition, the time between uses could be quite large; during the time spent in fixed-wing flight, the batteries could be recharged, ready for the next use. This would change the constraint on energy storage requirement; instead of the energy used by the motors over the whole mission, only the maximum energy spent in a **single** use of the motors would be the constraint for battery mass.

This maximum occurs during the supply delivery mission, Figure 16 in appendix D shows this slice of time taken from the larger mission profile. In this section, REACH must: transition from fixed-wing to VTOL, land, release the payload, take off, and transition back into fixed-wing flight. The area under this plot was then used to find the required battery mass to perform this section. A large safety factor of 2.5 was applied; this was due to the nature of the mission, the environment that REACH will be attempting to land in is unknown and potentially dynamic; a large buffer is desireable to allow for longer times to hover and land, should it be needed.

Using this battery mass, another plot could be created, one of battery percentage remaining, where during times of fixed-wing flight, a constant 1 kW (minus the power consumption of the other systems [12][48]) is given back to the batteries, recharging them, as shown in figure 12.

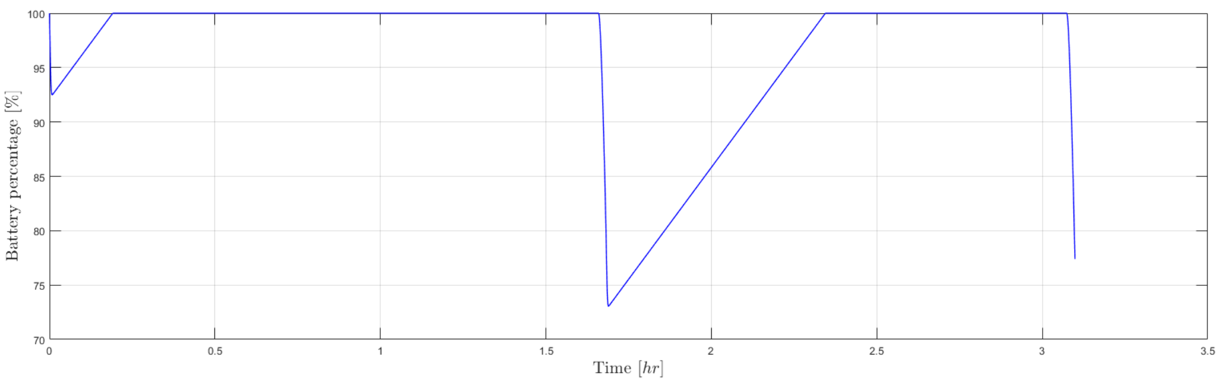

Figure 12. Battery percentage during supply delivery mission

Note that this power from the generator will be coming from burning Jet-A, so an important consideration was the added mass of jet-A that would be needed to recharge the batteries. However, it quickly became apparent that due to the two orders of magnitude difference in energy density between the batteries and Jet-A, any extra mass of Jet-A is insignificant compared to the battery mass that would be saved by recharging in flight. Such, this idea was taken forward into the final design.

3.4 Final Values and performance

Table 4 displays the final values of mass of Jet-A and batteries needed for the two main mission profiles.

Energy source	**Supply delivery** $[kg]$	**Long endurance** $[kg]$
Jet-A	6.5	36.5
Battery	5.4	5.4

Table 4. final values of mass for jet-A and batteries

The 30 kg difference of Jet-A between the two mission profiles is simply making up for the mass that is lost by having a lighter payload, as discussed in Section 3.1. 36.5 kg is more than enough fuel to complete the long endurance mission (only 18.15 kg needed) with its given parameters, so begs the question: what would the maximum endurance be on a full tank? The process of analysing the mission profile was repeated, except this time, loiter time was used as an independent variable, and finding the mass of fuel burnt; this provided a maximum loiter time of **26.6 hours** while still being able to perform the rest of the mission profile.

4 Conclusion, reflection, and improvements

Concluding this report, it was found that a UAS could be designed that met and exceeded expectations. REACH will be able to effectively mitigate the challenges faced by emergency services during disaster scenarios, and aid in the saving of lives.

Communication with academic supervisors proved to be extremely valuable over the course of the project. Speaking to someone who was involved in the project, but not directly working on its development, allowed for a ”third party” perspective; over the course of the project, the team often found itself ”stuck” in a certain idea of how something needs to be done, but upon consulting an academic supervisor, a more reasonable alternative would be suggested that had not previously been considered. This ultimately helped to design a UAS that is truly fit for purpose.

Looking to possible improvements for future iterations of the UAS, there were some ideas that were rejected due to time constraints; such as an option of performing the shorter missions fully electric, reducing environmental impact. This was an idea that may be feasible, and especially moreso when looking to the future as batteries become more energy dense. Further, optimisation of parameters was not done on a large scale; the foundation for it was made, as the author of this report ensured that code was fully parameterised, reading off a centralised ”parameter spreadsheet” that was kept up to date by the whole team. This meant that there was potential for the whole design process to be wrapped in an optimisation algorithm for the parameters, but was not done also due to time constraints.

It was noticed that Jet-A is very energy dense as a fuel, and this was shown through a possible endurance significantly above the requirement; in hindsight, perhaps a Wankel engine would have been better suited for REACH, since due to the payload mass problem, extra fuel had to be carried anyways, so fuel efficiency became less of an important factor. Further, due to Wankel engines versatility in fuels, an investigation could have been done looking into alternative fuels, that would still allow the UAS to meet requirements, but perhaps be cheaper or better for the environment, such as biofuels.

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Appendix

A Powerplant selection choices

Type	Choice	Important specifications
Engine	Hirth 4202HF	- Most efficient flight at SFC = 0.4 $kg/kWhr$ - Max power output 11 kW - 11.5kg mass [47] - 0.8 kg, 1 kW generator included
Motor	T-Motor H140 KV95	- Max 13.2 kW output, meeting take-off requirements [4] - 2.57 kg mass
Battery	Amprius SIMAXX	- Energy density 450 Whr/kg - mass/volume density 2600 kg/m^3

Table 5. Final powerplant selection

B Wankel rotary engines

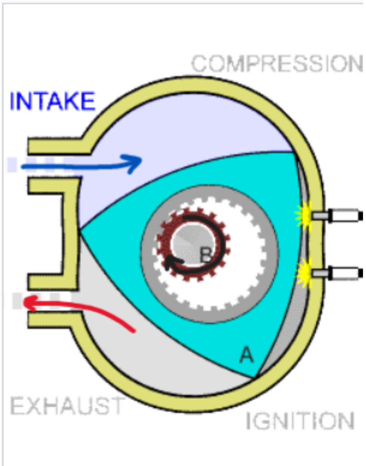

Figure 13. 2D diagram of a Wankel engine [42]

C Transition velocity/time relation derivation

To start, the forces in the horizontal direction, propeller thrust, T_{prop}, and drag, D, are used with Newton's second law to derive a relation with horizontal momentum.

$$T_{prop} - D = \frac{d}{dt}(mV) \tag{12}$$

Definitions for power (equation 2) and drag can be substituted in using air density, ρ, velocity, V, aircraft reference area, S_{ref}, and coefficient of drag, C_d; since fuel is burning, mass cannot be assumed constant, so product rule separates the derivative.

$$\frac{P_{engine}\,\eta_p}{V} - \frac{1}{2}\rho V^2 S_{ref} C_d = m\frac{dV}{dt} + V\frac{dm}{dt} \tag{13}$$

Engine properties can be used to find $\frac{dm}{dt}$; the rate of change of mass is equal to the fuel consumption (kg/s), as that is the only contribution to mass change. and then an unbounded integration with respect to time to find $m(t)$, with initial mass, m_0.

$$\frac{dm}{dt} = -SFC \cdot P_{engine} \tag{14}$$

$$m = m_0 - (SFC \cdot P_{engine})\, t \tag{15}$$

substituting (14) and (15) into (13) forms the differential equation relating velocity and time through constant parameters:

$$\frac{P_{engine}\,\eta_p}{V} - \frac{1}{2}\rho V^2 S_{ref} C_d \;=\; (m_0 - (SFC \cdot P_{engine})\, t\,)\,\frac{dV}{dt} + V\,(-\mathrm{SFC} \cdot P_{engine}) \tag{16}$$

D Mission profile power requirements

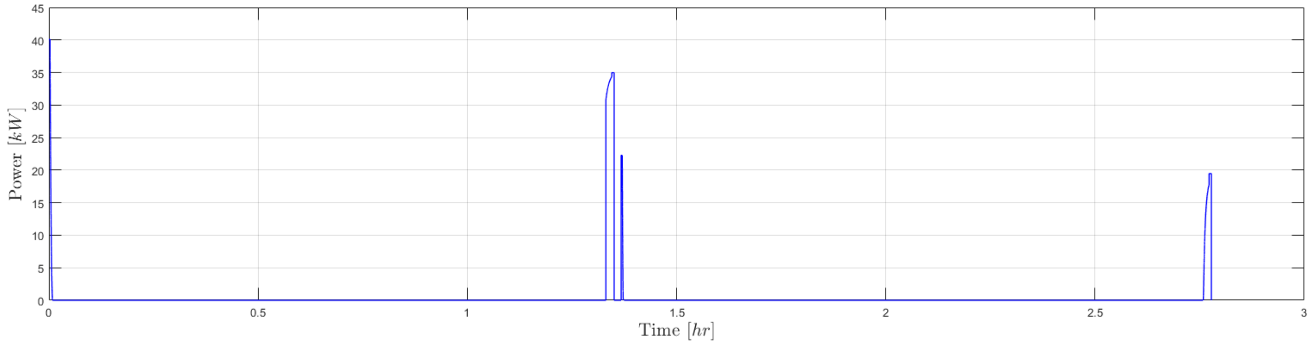

Figure 14. Power output of the motors during supply delivery mission

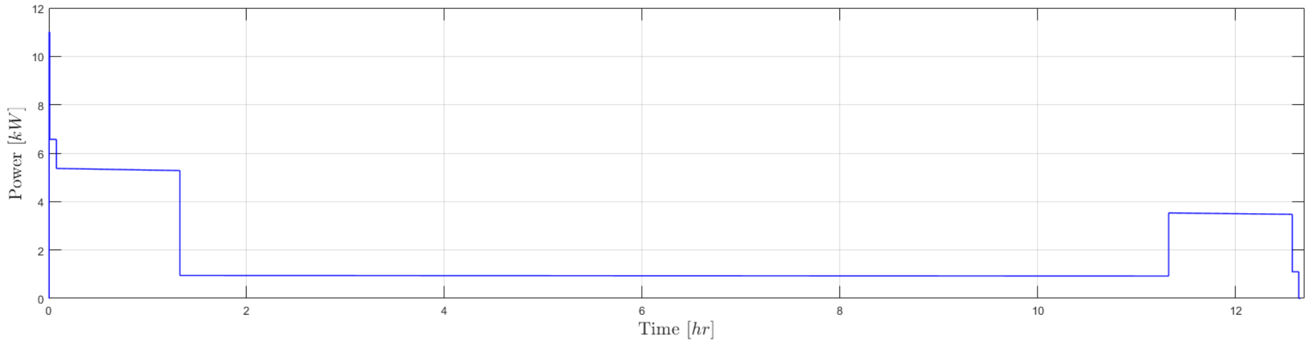

Figure 15. Power output of engine during long endurance mission

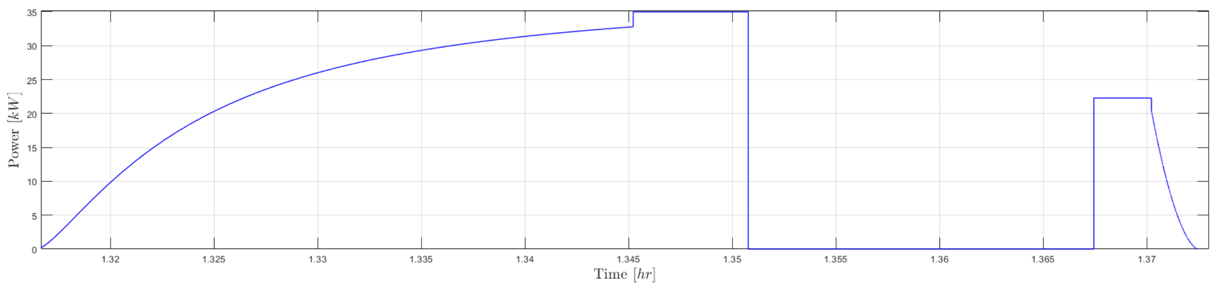

Figure 16. Constraining section for battery mass